SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE CODE: ITA 06 COURSE NAME: MACHINE LEARNING

COURSE OUTCOME COVERED

CO3: Bayes Theorem – Concept Learning – Maximum Likelihood – Minimum Description Length Principle – Bayes Optimal Classifier – Gibbs Algorithm – Naïve Bayes Classifier – Bayesian Belief Network – EM Algorithm – Probability Learning – Sample Complexity – Finite and Infinite Hypothesis Spaces – Mistake Bound Model, (BL 6)

Assessment Tool Weightage: 20%

QUESTIONS: 03 **Total Marks:25**

Annexure A

Case Study

Code of Conduct:

I ______________________(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Criteria	*Conceptual Understanding*	*Accuracy of Answers*	*Application of Knowledge*	*Completeness of Response*	*Clarity & Presentation*	*Total*
Weightage	*30%*	*20%*	*20%*	*15%*	*15%*	
Q1						
Q2						

Signature of the student: ***Total Marks:***

Annexure A

Short Answer Test

1, A hospital aims to develop an intelligent system to diagnose diseases based on patient symptoms and medical history. Using concepts such as Bayes' Theorem, Naïve Bayes classifiers, and Bayesian Belief Networks, design a probabilistic model for diagnosis. Analyze how uncertainty is handled and how conditional dependencies between symptoms influence predictions. Discuss the role of probability learning and evaluate the effectiveness of your approach with respect to real-world constraints.

2. An organization wants to build a spam detection system for emails using machine learning techniques. Using Naïve Bayes and Bayes Optimal Classifier concepts, design a classification system. Compare Maximum Likelihood estimation with Minimum Description Length (MDL) principle in feature selection. Analyze system performance, discuss trade-offs in model complexity, and evaluate how sample size and hypothesis space affect classification accuracy.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	**30**	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	**20**	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	**20**	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	**15**	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	**15**	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

Q1. Intelligent Disease Diagnosis Using Probabilistic Models

Introduction

A hospital wants to develop an intelligent system that can diagnose diseases using patient symptoms and medical history. A probabilistic model can be designed using **Bayes' Theorem, Naïve Bayes Classifier, and Bayesian Belief Network (BBN)**. These techniques help the system make predictions even when the available information is uncertain or incomplete.

Proposed Model

The system takes the following inputs:

- Patient symptoms
- Medical history
- Age and other relevant patient information
- Test results

The output is the probability of different diseases.

Bayes' Theorem is used to calculate the probability of a disease given the observed symptoms:

P (Disease | Symptoms) = [P (Symptoms | Disease) × P(Disease)] / P(Symptoms)

Here:

- **P(Disease)** is the prior probability.
- **P(Symptoms | Disease)** represents the likelihood.
- **P(Disease | Symptoms)** is the posterior probability.

The disease with the highest posterior probability can be considered the most likely diagnosis.

Naïve Bayes Classifier

Naïve Bayes assumes that symptoms are conditionally independent given a disease. For example, symptoms such as fever, cough, and fatigue can be used as features.

The classifier calculates the probability of each disease based on the observed symptoms and selects the disease with the highest probability.

Bayesian Belief Network

A Bayesian Belief Network represents relationships between medical variables using a graphical structure.

For example:

Disease → **Fever**

Disease → **Cough**

Disease → **Fatigue**

Disease → Blood Test Result

This allows dependencies between symptoms and medical conditions to be represented. Unlike the simple independence assumption of Naïve Bayes, a BBN can model relationships between different variables.

Handling Uncertainty

Medical diagnosis involves uncertainty because:

- Symptoms can occur in multiple diseases.
- Patients may not report all symptoms.
- Medical test results may be uncertain.
- Patient history may be incomplete.

Probability values allow the system to express uncertainty rather than producing only a fixed yes/no answer.

Probability Learning

The probabilities can be learned from historical patient data. As more patient records become available, the system can update its probability estimates and improve its predictions.

Real-World Constraints

The system should consider:

- Quality and availability of medical data
- Missing patient information
- Sensor and test errors
- Privacy of patient information
- Computational requirements
- Need for explainable predictions

The system should support doctors rather than completely replace professional medical judgment.

Conclusion

A combination of Bayes' Theorem, Naïve Bayes, and Bayesian Belief Networks can provide an effective probabilistic diagnosis system. Bayes' Theorem handles uncertainty, Naïve Bayes provides efficient classification, and Bayesian Belief Networks represent dependencies between medical variables. Probability learning allows the model to improve as more data becomes available.

Q2. Spam Detection Using Naïve Bayes and Bayes Optimal Classifier

Introduction

An organization wants to develop an email spam detection system. The system classifies incoming emails as **Spam** or **Not Spam** using probabilistic machine learning techniques. Naïve Bayes and the Bayes Optimal Classifier can be used to make these predictions.

Proposed System

The system processes an incoming email and extracts features such as:

- Words in the email
- Number of links
- Presence of suspicious words
- Sender information
- Frequency of special characters
- Email length

These features are used to calculate the probability that an email belongs to the spam class.

Naïve Bayes Classification

Naïve Bayes calculates:

P(Spam | Features)

and

P(Not Spam | Features)

using Bayes' Theorem.

The system compares both probabilities:

- If **P(Spam | Features)** is higher → classify as **Spam**.
- If **P(Not Spam | Features)** is higher → classify as **Not Spam**.

Naïve Bayes is suitable for spam detection because it is simple, fast, and works well with high-dimensional text data.

Bayes Optimal Classifier

The Bayes Optimal Classifier considers all possible hypotheses and makes a prediction based on their posterior probabilities.

Instead of depending on only one hypothesis, it considers the uncertainty among multiple possible models. This can provide better theoretical classification performance when accurate probability estimates are available.

Maximum Likelihood Estimation vs MDL

Maximum Likelihood Estimation (MLE):

MLE selects parameters that maximize the likelihood of observing the training data.

For spam detection, it can estimate how frequently particular words or features occur in spam and non-spam emails.

Minimum Description Length (MDL):

MDL selects a model by considering both:

- The complexity of the model
- How well the model describes the data

MDL attempts to avoid unnecessarily complex models and can help reduce overfitting.

Comparison

MLE	MDL
Focuses on maximizing likelihood	Focuses on model and data description length
Can produce complex models	Encourages simpler models
May overfit with limited data	Helps control overfitting
Mainly concerned with data fit	Balances fit and model complexity

Effect of Sample Size

A larger training dataset generally provides better estimates of probabilities. With a small sample size, the system may incorrectly estimate the probability of words or features.

Therefore:

More training data → Better probability estimates → Potentially better classification accuracy

However, the quality of the training data is also important.

Effect of Hypothesis Space

A **hypothesis space** represents the possible models that can be considered.

A very large hypothesis space provides more possible models but increases computational complexity and may increase the risk of overfitting. A smaller and appropriate hypothesis space can make learning faster and easier to generalize.

Performance and Trade-Offs

The spam detection system can be evaluated using:

- Accuracy
- Precision
- Recall
- F1-score
- False-positive rate
- False-negative rate

For spam detection, reducing **false positives** is important because legitimate emails should not be incorrectly classified as spam.

Conclusion

Naïve Bayes provides a fast and effective approach for spam classification, while the Bayes Optimal Classifier provides a probabilistic framework for considering multiple hypotheses. MLE focuses on fitting the observed data, whereas MDL balances model complexity with data fitting. Sample size and hypothesis-space size strongly influence classification accuracy and generalization.

These two are good choices to write for the assessment because they cover the main CO3 concepts and satisfy the rubric's requirements for understanding, application, completeness, and clarity.